

PSG college of Technology
Department of Computer Applications
23MX18 Web Application Development
Worksheet 12

Scenario

You are developing a registration module for a web-based student portal. One of the key requirements is to ensure that users create **strong passwords** to secure their accounts. The password must be validated before submission to the server.

Your task is to **design and implement a web page using HTML, CSS, and JavaScript** that checks the validity and strength of a password entered by the user, following the specifications below.

Functional Requirements:

1. Input Field:

- A text field (type = password) for the user to enter a password.

2. Validation Rules:

The password must satisfy **all** the following conditions:

- Minimum length: **8 characters**
- Must contain at least **one uppercase** letter (A–Z)
- Must contain at least **one lowercase** letter (a–z)
- Must contain at least **one numeric digit** (0–9)
- Must contain at least **one special character** (@, #, \$, %, &, *)
- Must **not contain spaces**

3. Output Message:

- If all conditions are met → display the message in **green**:
"Password is strong and valid."

- If not → display the message in **red**:
"Password is invalid. Please follow the rules below."
- Additionally, show a list of **missing requirements** below the message (for example, "Missing special character", "Too short", etc.)

4. Styling Requirements:

- Use **Arial, sans-serif** for all text.
- Center the form on the page.
- Highlight valid criteria in green and invalid ones in red as the user types.

5. Implementation Constraint:

- The validation must be done **entirely using JavaScript** (no HTML5 pattern attribute).
- The form must **not be submitted** if the password is invalid.

Expected Output Example:

Enter Password: Pass@12

Password is invalid. Please follow the rules below.

Must be at least 8 characters

Must contain one lowercase letter

If valid:

Enter Password: Pass@123

Password is strong and valid.